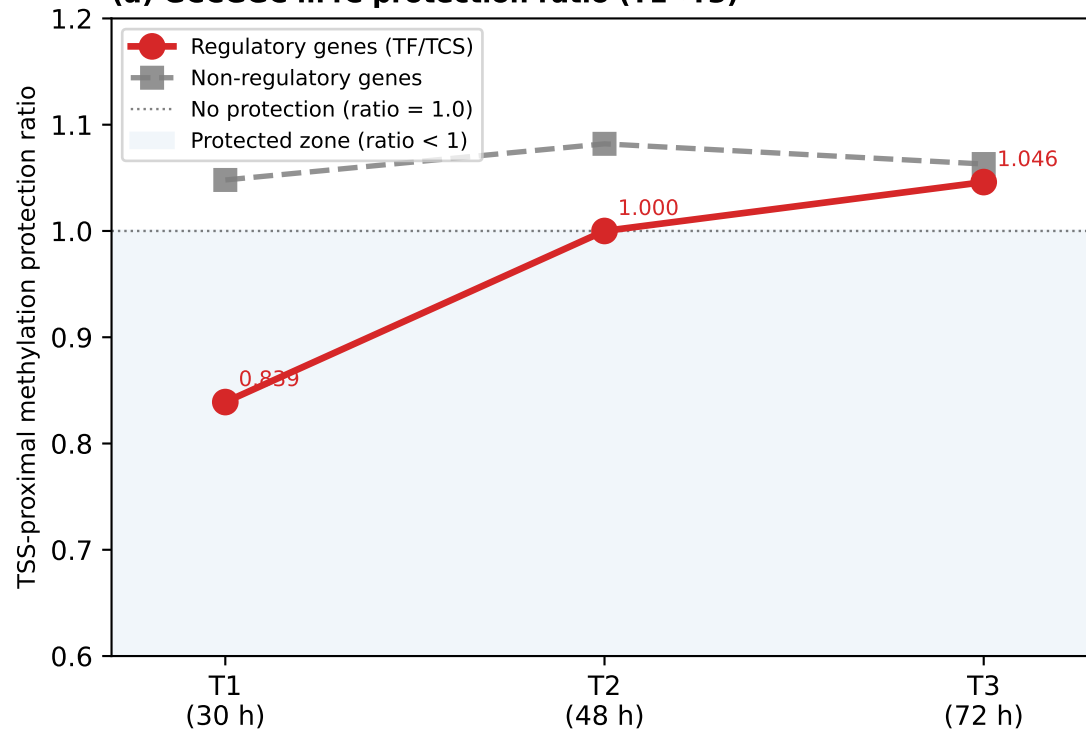
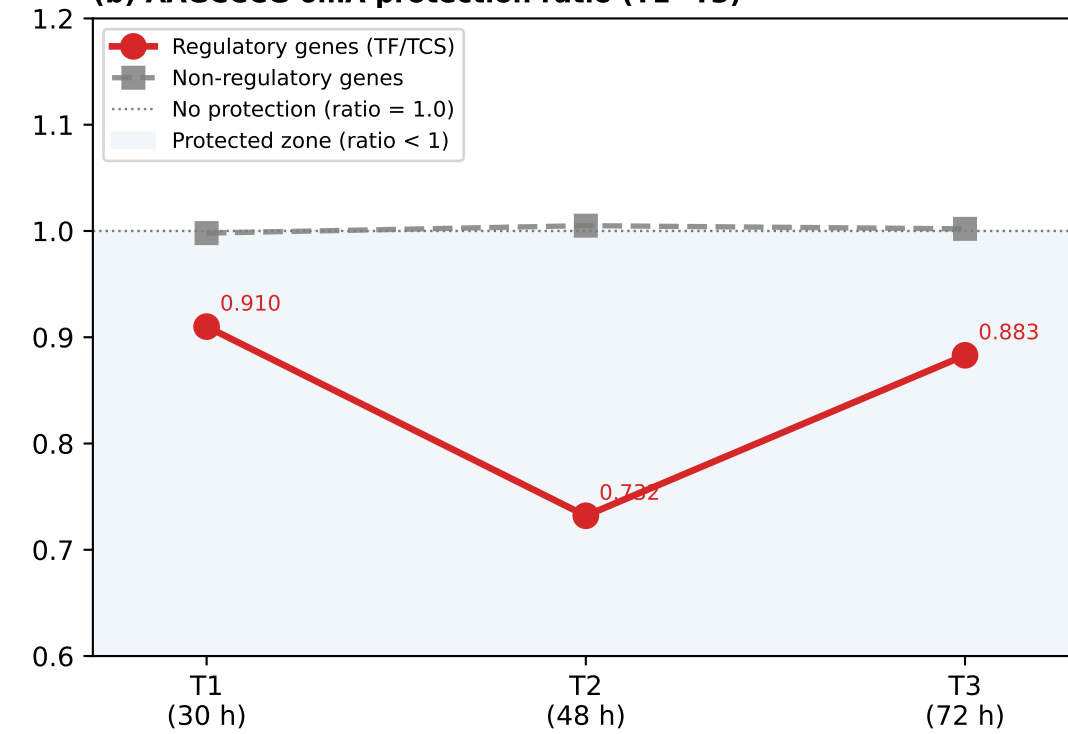


Supplementary Figure S_ | Temporal dynamics of TSS-proximal methylation protection zones

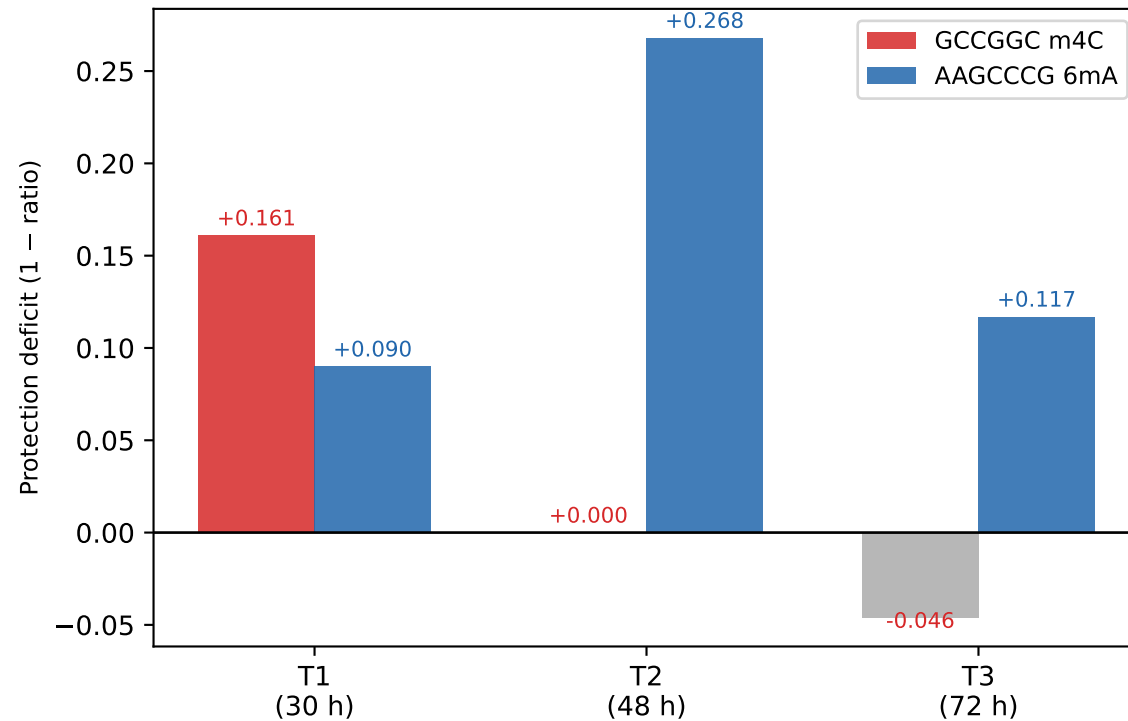
(a) GCCGGC m4C protection ratio (T1→T3)



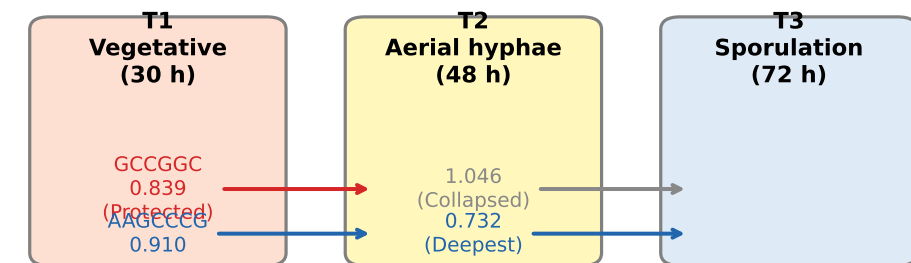
(b) AAGCCCG 6mA protection ratio (T1→T3)



(c) Regulatory gene protection deficit by timepoint (positive = protected, negative = exposed)



(d) Protection zone dynamics — model summary



*4mC protection COLLAPSES at T2→T3 (0.839→1.046)
6mA protection DEEPENS at T1→T2 (0.910→0.732)
Suggesting independent regulatory roles for each modification*